

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Mid Semester Exam  
Duration: 1 Hour

Semester :Summer 2022  
Full Marks: 30

**CSE 221: Algorithms**

**Set: A**

Answer the following questions.  
Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

1. a. **Calculate** the time complexity of the following function: **5**  
CO7

```
worstCase(n):  
    int i, j, k, a, b, sum  
    for ( i = 0; i < n; i = i + 3)  
        for ( j = n; j >= 1; j = j / 5)  
            for ( k = 1; k <= n; k = k * 5)  
                sum = a + b
```

- b. **Calculate** the time complexity of the following recurrence relation. **5**  
CO7 [Any method is acceptable as long as steps are shown]

$$T(n) = 2T(n/4) + \sqrt{n}$$

2. Renowned Progressive Rock band 'Porcupine Tree' released an album called 'Closure/Continuation' after about 13 years. Now as a Progressive Rock Music fan you are going to listen to the tracks of the album but in the order of their Youtube views (highest one at first, lowest one at last) instead of the order of the album tracklist.

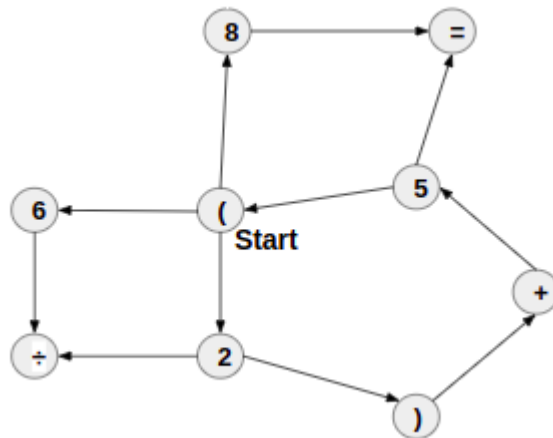
You have chosen an Algorithm to order them as per your preference. If multiple tracks have the same views, you are going to listen to any of them the Algorithm puts first in the list after ordering. This algorithm also solves your concern about the issues regarding space that your device is facing. Porcupine Tree made the fans wait for 13 years for a new album and so you think you have the patience to wait as much time as the Algorithm may take to order the tracks.

The following table contains the list of the tracks and their Youtube views (in thousand) :

| Track | Harridan | Of The New Day | Rats Return | Dignity | Herd Culling | Walk the Plank | Chimera's Wreck | Population Three |
|-------|----------|----------------|-------------|---------|--------------|----------------|-----------------|------------------|
| Views | 15       | 8              | 11          | 112     | 33           | 39             | 88              | 41               |

- a. **Specify** the name of the algorithm you have chosen and **simulate** the Algorithm to order the tracks of the album as per your preference. Show your workings in detail. **7**  
CO2
- b. Suppose you want to run the algorithm again on the ordered list. **3**  
CO5 **Determine** the run-time complexity of the algorithm in this scenario.

3. Consider the graph shown below:



Bill says he has found a valid mathematical equation while running DFS from the node denoted by '('. According to him, the equation is:

$$(6 \div 2) + 5 = 8$$

- |     |                                                                                                                                      |          |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|----------|
| a.  | <b>Compute</b> the number of edges this DFS tree (of the equation) contains.                                                         | <b>1</b> |
| CO3 |                                                                                                                                      |          |
| b.  | Is what Bill said right i.e is the equation achievable by running such a DFS?                                                        | <b>5</b> |
| CO3 | <b>Validate</b> it by showing steps. (Neighbor/Edge selection should be done according to the necessity of the verification process) |          |
| c.  | <b>Classify</b> the edges of the graph into tree edge, back edge and forward edge while running                                      | <b>4</b> |
| CO2 | DFS from the node denoted by '('.                                                                                                    |          |

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**CSE 221: Algorithms**

**Set: B**

Answer the following questions.  
Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

1. a. **Calculate** the time complexity of the following function: 5  
CO7

```
worstCase(n):  
    int i, j, k, a, b, sum  
    for ( i = 1; i < n; i = i + 3)  
        for ( j = n; j >= 1; j = j - 2)  
            for ( k = n; k >= 1; k = k / 4)  
                sum = a + b
```

- b. **Calculate** the time complexity of the following recurrence relation. 5  
CO7 [Any method is acceptable as long as steps are shown]

$$T(n) = 2T(n/2) + \frac{1}{n}$$

2. Renowned Progressive Rock band 'Porcupine Tree' released an album called 'Closure/Continuation' after about 13 years. Now as a Progressive Rock Music fan you are going to listen to the tracks of the album but in the order of their Youtube views (highest one at first, lowest one at last) instead of the order of the album tracklist.

You have chosen an Algorithm to order them as per your preference. If multiple tracks have the same views, you are going to listen to any of them the Algorithm puts first in the list after ordering. This algorithm also solves your concern about the issues regarding space that your device is facing. Porcupine Tree made the fans wait for 13 years for a new album and so you think you have the patience to wait as much time as the Algorithm may take to order the tracks.

The following table contains the list of the tracks and their Youtube views (in thousand) :

| Track | Harridan | Of The New Day | Rats Return | Dignity | Herd Culling | Walk the Plank | Chimera's Wreck | Population Three |
|-------|----------|----------------|-------------|---------|--------------|----------------|-----------------|------------------|
| Views | 17       | 10             | 15          | 88      | 35           | 41             | 90              | 38               |

- a. **Specify** the name of the algorithm you have chosen and **simulate** the Algorithm to order the tracks of the album as per your preference. Show your workings in detail. 7  
CO2
- b. Suppose you want to run the algorithm again on the ordered list. 3  
CO5 **Determine** the run-time complexity of the algorithm in this scenario.

3. Bill has this weird characteristic of playing with different types of graphs. This time he comes up with an undirected simple graph of 10 nodes. Nodes are labeled from 3 to 12. Some of the edges are:

(5,6), (5,10), (6,11), (7,11). He created other edges following two rules.

- a. All the nodes that are labeled with a number which is a multiple of 3 have edges among them.
- b. All the nodes that are labeled with a number which is a multiple of 4 have edges among them.

- |     |                                                                                                                                                                             |          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| a.  | <b>Draw</b> the graph to show all the edges.                                                                                                                                | <b>1</b> |
| CO3 |                                                                                                                                                                             |          |
| b.  | Bill says, there are at least four triangles in the graph. Do you agree?                                                                                                    | <b>2</b> |
| CO3 | <b>Support</b> your answer by showing the nodes which form these triangles.                                                                                                 |          |
| c.  | <b>Simulate</b> a BFS algorithm on the graph to find the shortest distance from Node 3 to all others nodes.                                                                 | <b>6</b> |
| CO2 |                                                                                                                                                                             |          |
| d.  | Bill told you to keep on adding edges between the nodes according to your wish, keeping the graph simple (without adding multiple edges between any two nodes, self edges). | <b>1</b> |
| CO3 | <b>Compute</b> the number of more edges that you can add.                                                                                                                   |          |

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Mid Semester Exam  
 Duration: 1 Hour 30 Minutes

Semester: Summer 2023  
 Full Marks: 40

**CSE 221: Algorithms**

Answer the following questions.  
 Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

- 1 a. Find the time-complexity of the following task in terms of number of students. 03  
 CO2

You are given a student attendance sheet. Each student has a unique integer ID. You have to count the number of students having an even number as ID. The list is sorted but the IDs are not necessarily consecutive. So you check each ID one by one.

- b. Write the asymptotic time complexity of the following code snippet. Show your works/reasoning. 02  
 CO2

```
1. for i in range (1,n)
2.     j= 1
3.     while j < i*i
4.         j= j+1
```

- c. Express the following running time  $T(n)$  with an asymptotic bound. 05  
 CO2

$$T(n) = 8T\left(\frac{n}{4}\right) + n\sqrt{n}$$

Any method is acceptable as long as you show calculations.

- 2 a. You are given an array containing  $N$  distinct integers in ascending order. You have to find out the first missing number from the increasing sequence. 01  
 CO3
- For example: [3, 4, 5, 6, 9, 10]. The first missing number in this sequence is 7.  
 [Explanation : If  $A_i$  and  $A_{i+1}$  are two consecutive integers from the list and their difference ( $A_{i+1} - A_i$ ) is more than 1, then we say  $A_i+1$  is a missing number.]

Now that you have understood the problem completely, you are trying to solve it efficiently. Your friend, Jack, said that by iterating through all the numbers from the beginning, you can find the first missing number in linear time  $[O(N)]$ . But, you need a more efficient one. Now, **answer** the questions below.

- a) If  $N = 9$  and the maximum number (obviously the last-indexed one) in the array is 15, which one among 6, 8, 10 might be the first number (obviously the minimum one) in the array? 01
- b) Explain why you discarded the other two numbers in question no. (a). 01

- c) Assume  $N = 7$  and the minimum and maximum numbers are 2 and 12 respectively. You found the middle-index value to be 5. On which side (left/right) of 5, do you expect to find the first missing number? Only mention 'left' or 'right'. **01**
- d) Explain why you discarded the other half of the array in question no. (c). **01**
- e) Assume  $N = 7$  and the minimum and maximum numbers are 8 and 19 respectively. The missing elements are 11, 13, 14, 17, 18. What will be the value of the middle-index element? Determine the absolute difference between the middle-index value and the maximum number. **02**

- b. CO1** **Simulate** the Quick Sort algorithm by showing the steps on the following array, always choosing the last element as the pivot element. You can partition the array at each step choosing any suitable algorithm, *but you do not need to show the partitioning steps.* **04**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 51 | 70 | 28 | 12 | 75 | 62 | 88 | 43 |
|----|----|----|----|----|----|----|----|

- 3 CO1** Imagine you're a farmer named Sam who grows different types of vegetables in several continuous fields. Each field can grow a specific type of vegetable, and based on the market value of that vegetable, each field will bring a certain profit (positive integer) or loss (negative integer). The profit or loss is estimated and noted for each field.

Sam can start farming from any field, but once he starts, he must continue farming the next fields in sequence without skipping any, until he decides to stop, because his tractor can only move to the next adjacent field and cannot skip fields.

Sam needs to choose which sequence of fields to farm to maximize his profits. The following array represents the estimated profits or losses for ten consecutive fields on Sam's farm, based on the types of vegetables that can be grown in each field and their respective market values.

[4, -18, 16, -14, 12, -1, 3, 1, -20, 15]

- a) Can you help Sam decide which sequence of fields to farm for maximum profit using an efficient algorithm? What would the profit amount be? **Show** a simulation of your proposed algorithm. You must show which sequence of fields he needs to select in order to achieve this maximum profit. **06**
- b) **Calculate** the time complexity of your algorithm using proper mathematical logic. An efficient algorithm should have time complexity less than or equal to  $O(N * \log(N))$  where  $N$  is the number of fields. **04**

- 4 CO1** The cities in Italy are connected in such a way that their adjacency list representation looks as follows-

Rome: Milan, Venice, Florence  
 Milan: Rome, Florence, Bologna, Venice  
 Florence: Milan, Naples, Rome, Bologna, Venice  
 Venice: Rome, Milan, Florence  
 Naples: Florence, Bologna, Genoa  
 Turin: Genoa  
 Bologna: Naples, Genoa, Milan, Florence  
 Genoa: Turin, Bologna, Naples  
 Palermo: Catania, Syracuse  
 Catania: Palermo, Syracuse  
 Syracuse: Catania, Palermo

Suppose Rome suddenly gets affected with covid-19. The rest of the cities are non-affected. You are asked to find out the time steps the covid virus will take to affect all the non-affected cities if it takes one time step to travel from one city to another. So, one affected city will affect all of its neighbors in one time step.

- a) **Draw** the graph and create the adjacency matrix for the graph. **04**
- b) **Mention** the name of the graph traversal algorithm you think is suitable for solving the problem. **01**
- c) **Simulate** the suitable algorithm to find the time steps needed for each non affected city to be affected. Choose the smallest (in alphabetical order) vertex when there is a choice. **05**

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Semester: Summer 2023  
 Full Marks: 40

**CSE 221: Algorithms**

Answer the following questions.  
 Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

- 1 a. Find the time-complexity of the following task in terms of number of students. 03  
 CO2

You are given a student attendance sheet. Each student has a unique integer ID. You have to count the number of students having an ID which is divisible by 3. The list is sorted but the IDs are not necessarily consecutive. So you check each ID one by one.

- b. Write the asymptotic time complexity of the following code snippet. Show your works/reasoning. 02  
 CO2

```
1. for i in range (1,n)
2.     j= 1
3.     while j*j < i
4.         j= j+1
```

- c. Express the following running time  $T(n)$  with an asymptotic bound. 05  
 CO2

$$T(n) = 4T\left(\frac{n}{4}\right) + \sqrt{n}$$

Any method is acceptable as long as you show calculations.

- 2 a. You are given an array containing  $N$  distinct integers in descending order. You have to find out the first missing number from the decreasing sequence. 01  
 CO3
- For example: [10, 9, 6, 5, 4, 3]. The first missing number in this sequence is 8.  
 [Explanation : If  $A_i$  and  $A_{i+1}$  are two consecutive integers from the list and their difference ( $A_i - A_{i+1}$ ) is more than 1, then we say  $A_i - 1$  is a missing number.]

Now that you have understood the problem completely, you are trying to solve it efficiently. Your friend, Jack, said that by iterating through all the numbers from the beginning, you can find the first missing number in linear time  $[O(N)]$ . But, you need a more efficient one. Now, **answer** the questions below.

- a) If  $N = 9$  and the maximum number (obviously the first-indexed one) in the array is 17, which one among 8, 10, 12 might be the last number (obviously the minimum one) in the array? 01
- b) Explain why you discarded the other two numbers in question no. (a). 01

- c) Assume  $N = 7$  and the maximum and minimum numbers are 10 and 3 respectively. You found the middle-index value to be 6. On which side (left/right) of 6, do you expect to find the first missing number? Only mention 'left' or 'right'. 01
- d) Explain why you discarded the other half of the array in question no. (c). 01
- e) Assume  $N = 7$  and the maximum and minimum numbers are 23 and 12 respectively. The missing elements are 21, 20, 17, 16, 13. What will be the value of the middle-index element? Determine the absolute difference between the middle-index value and the minimum number. 02

**b. CO1** Draw the recursion tree of the Merge Sort algorithm while sorting the following array. Show the sorted lists at each step. 04

|    |    |    |    |    |   |    |    |
|----|----|----|----|----|---|----|----|
| 53 | 23 | 87 | 14 | 21 | 9 | 36 | 84 |
|----|----|----|----|----|---|----|----|

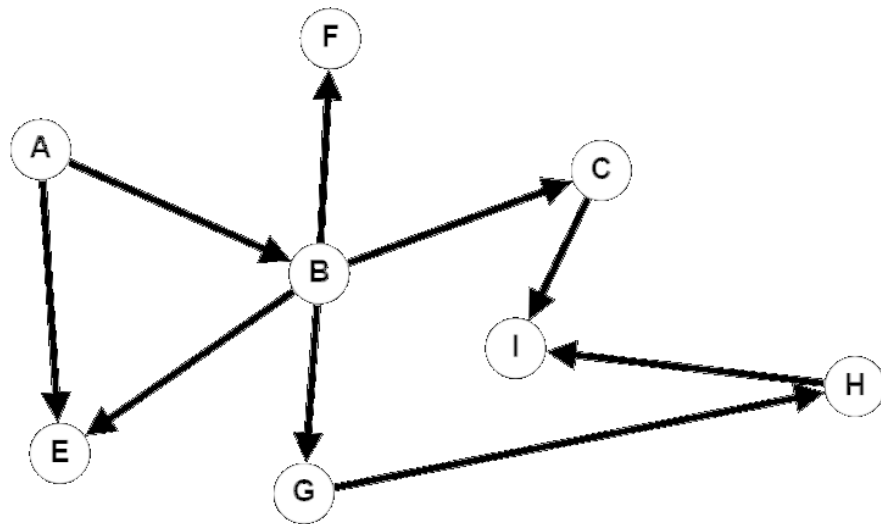
**3 CO1** John is an amateur poker player who has decided to try his luck at a local casino. The casino allows players to play in rounds, each leading to a win (positive integer) or a loss (negative integer). John keeps track of his net gain or loss after each round.

John knows that poker isn't just about luck—it's about making strategic decisions. And one of the critical decisions he has to make is knowing when to stop playing. However, the casino has a rule that once a player starts playing rounds, they must play the subsequent rounds in order without skipping any, until they decide to stop, as skipping rounds might disrupt the flow of the game and disadvantage other players.

With this in mind, John wants to find the optimal sequence of rounds where he can maximize his net winnings. The following array represents the net gain or loss John experiences in ten rounds of poker at the casino.

[8, -18, -8, 2, 16, -14, 2, -1, 3, 15]

- c) Find an efficient algorithm that will help John to figure out which sequence of rounds he should play to maximize his winnings. What would be the exact profit amount? **Show** a simulation of your proposed algorithm. You must show which sequence of rounds he needs to select in order to achieve this maximum profit. 06
- d) **Calculate** the time complexity of your algorithm using proper mathematical logic. An efficient algorithm should have time complexity less than or equal to  $O(N * \log(N))$  where  $N$  is the number of fields. 04



- (a) **Apply** the DFS algorithm on the following graph starting from vertex **A**. Determine start time, finish time, and parent for each of the vertices. Choose the smallest (in alphabetical order) vertex when there is a choice. **06**
- (b) For each edge, **determine** whether it is a tree edge, forward edge, back edge or cross edge. **04**

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Mid Semester Exam  
Duration: 1 Hour 20 Minutes

Semester: Summer 2024  
Full Marks: 35

**CSE 221: Algorithms**

Answer the following questions.  
Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

- 1 a. Analyze the code snippet and answer the following questions:  
CO2

05

```
int func(int n) {
    if (n <= 1) {
        return 1;
    } else {
        int a = 0;
        for (int i = 0; i < 3; i++) {
            a += func(n / 2);
        }
        int b = func2(n);
        return a + b;
    }
}

int func2(int n) {
    int c = 0;
    for (int k = n; k >= 1; k -= 4) {
        for (int m = 0; m < n; m += 2) {
            c += k + m;
        }
    }
    return c;
}
```

- i. Write the recurrence relation of the following function "func".
- ii. Calculate the time complexity of the recurrence relation using any of the methods.

- b. For each of these pairs of functions ( $f(n)$ ,  $g(n)$ ), identify whether  $f(n) = O(g(n))$  or  $f(n) = \Omega(g(n))$ . 03

CO2

Show your calculation logic in the script and **Tick only the most specific one** of the two options in each case in the question paper. *Marks for this question depend on the calculation shown in your scripts.*

| $f(n)$                       | $g(n)$          | $f(n) = O(g(n))$         | $f(n) = \Omega(g(n))$    |
|------------------------------|-----------------|--------------------------|--------------------------|
| $3^n$                        | $2^n \cdot n^4$ | <input type="checkbox"/> | <input type="checkbox"/> |
| $(\sqrt{n} + n)(30\sqrt{n})$ | $n^2$           | <input type="checkbox"/> | <input type="checkbox"/> |

- c. Consider the recurrence relation  $T(n) = T(\sqrt{n}) + O(n)$ . Can we solve this using the Master Theorem? **Explain** whether the structure of this recurrence fits the requirements of the Master Theorem. 02

CO2

- 2 a. Your friend gave you an integer (key) and asked you to find its integer square root without using built-in functions like `sqrt()` or exponentiation. For example, if the input is 25, the output should be 5 because  $5^2 = 25$ . If the square root is not an integer, return the floor value of the actual square root. For example, for an input of 10, the output should be 3.

Your friend's initial approach is to use linear search (time complexity  $O(n)$ ) as shown below.

```
def LinearSearchToFindSquareRoot(key):
    result = -1
    for i in range(1, key, 1):
        if i * i <= key:
            result = i
    return result
```

Now to find the square root he is working on a possible solution **P**, which will result in time complexity  $O(\sqrt{n})$ . As an algorithm student, you propose another algorithm **R** within time complexity  $O(\log_2 N)$ , which is much faster than  $O(\sqrt{n})$ .

- i. Present the **solution your friend** was trying to achieve (**Algorithm P**) with pseudocode/programmable code/step-by-step logical instructions. 03
- ii. Present your proposed solution (**Algorithm R**) with pseudocode/programmable code/step-by-step logical instructions. 04

CO2

CO3

- 3 a. CO2 Imagine yourself helping Taylor Swift following her "Eras Tour first show." Taylor is curious about her performance and the groups of songs her audience connected most with them. She receives comments for every song on her setlist: some calm the audience down a bit (*and have a negative score*), while others get the crowd quite thrilled (*and have a positive score*).

Taylor can choose any song to start her performance, but she likes to keep the flow in order. So the next song she sings has to be the one right after she has chosen the starting song on the list, regardless of the public reactions.

Given a list of scores for each song (indicating how much they boost or calm the crowd), can you:

[-3, 7, 12, -8, -2, 83, -7, 4]

- i. Help Taylor determine which segment of consecutive songs she should perform at her next concerts to maximize audience interest. What is the total comment score of this segment? **Simulate** with proper steps. 05
- ii. **Explain** the time taken by your preferred method, particularly given the “ $n$ ” numbers of songs Taylor has in her setlist. Ideally, your method should not exceed  $O(n \lg n)$ . 03

- 4 The “Road Transport and Highways Division” of Bangladesh is working on a project where they want to establish six-lane highways among the 8 divisions of Bangladesh. Currently, there are 8 highways connecting these 8 divisions as given in the map below.

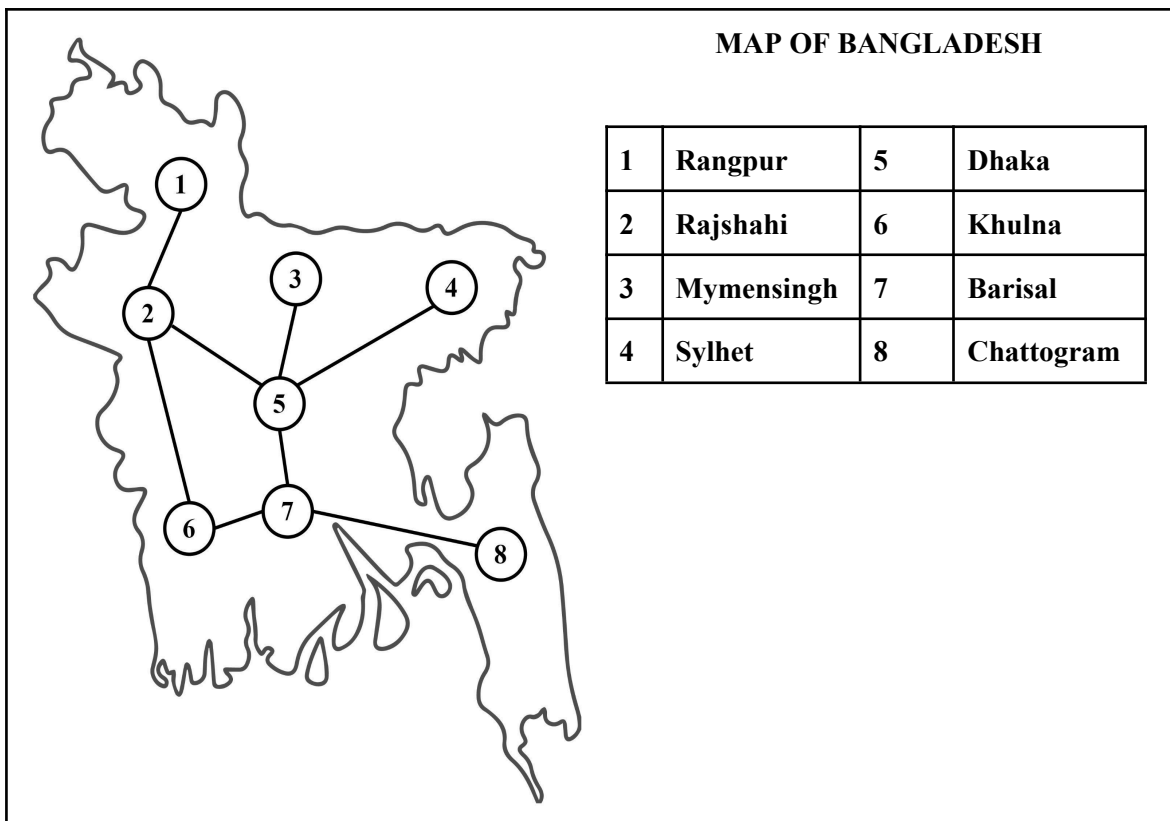


Fig 1: Graph For Question 4

- CO1 i. **Create** a graph representation for the given map so that it does not take more than  $O(|V| + |E|)$  space/memory. 02
- CO2 ii. **Determine** how many additional highways we need to make sure that every division is directly connected with each other. 02
- CO2 iii. In a meeting, the director looked at the map and wondered if the divisions of Bangladesh could be split into two groups (**A** and **B**) such that all highways directly connect locations in Group A to Group B, with no highways between locations within the same group. Using an appropriate data structure, simulate the algorithm that can determine if such a division is possible. 04
- CO2 iv. Based on the given graph in Fig 1, **explain** how you can find out if there is an odd number cycle present in the graph. An odd number cycle is a cycle that has an odd number of vertices. 02

For your convenience, Figure 1 is repeated below:

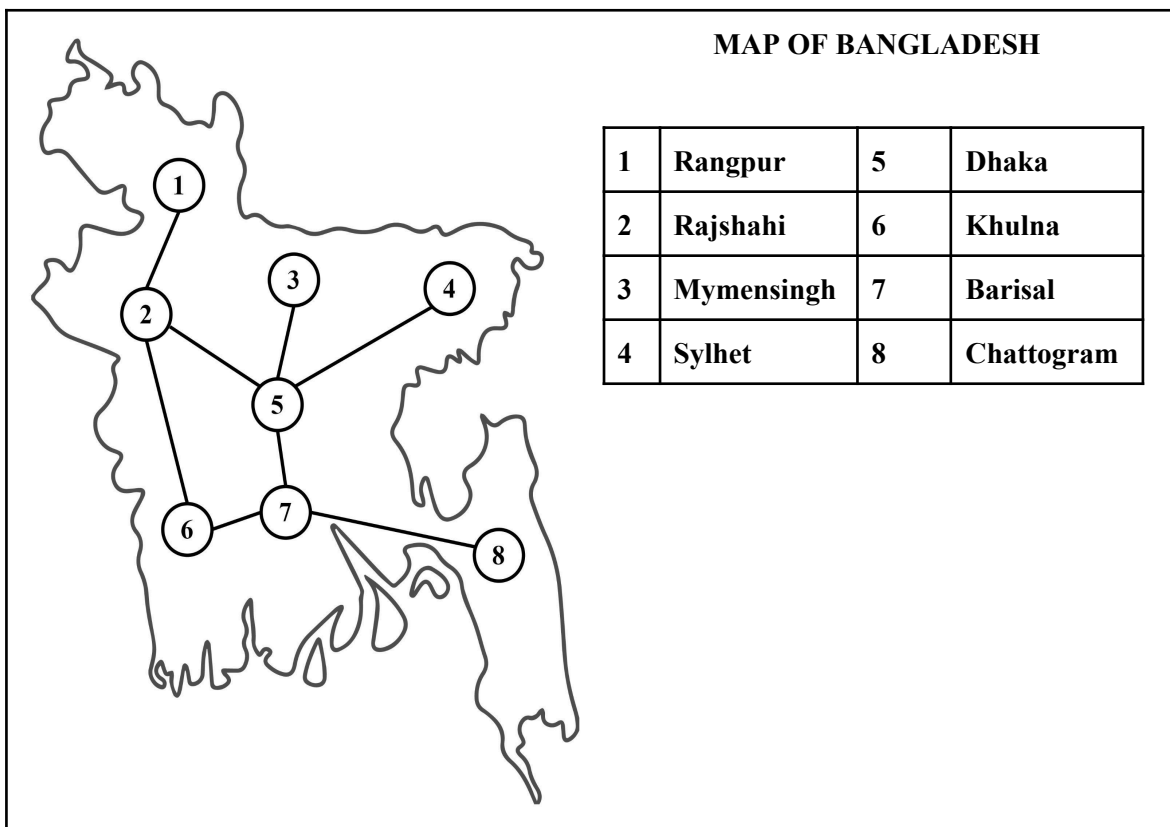


Fig 1: Graph For Question 4

**END OF QUESTIONS**

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

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Semester: Summer 2024  
Full Marks: 35

**CSE 221: Algorithms**

Answer the following questions.  
Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

- 1 a. Analyze the code snippet and answer the following questions:  
CO2

05

```
int func(int n) {
    if (n <= 1) {
        return 1;
    } else {
        int a = 0;
        for (int i = 0; i < 4; i++) {
            a += func(n / 4);
        }
        int b = func2(n);
        return a + b;
    }
}

int func2(int n) {
    int c = 0;
    for (int k = n; k >= 1; k -= 3) {
        for (int m = 0; m < n; m += 2) {
            c += k + m;
        }
    }
    return c;
}
```

- i. Write the recurrence relation of the following function "func".
- ii. Calculate the time complexity of the recurrence relation using any of the methods.

- b. For each of these pairs of functions ( $f(n)$ ,  $g(n)$ ), identify whether  $f(n) = O(g(n))$  or  $f(n) = \Omega(g(n))$ . 03

CO2

Show your calculation logic in the script and **Tick only the most specific one** of the two options in each case in the question paper. *Marks for this question depend on the calculation shown in your scripts.*

| $f(n)$                       | $g(n)$             | $f(n) = O(g(n))$         | $f(n) = \Omega(g(n))$    |
|------------------------------|--------------------|--------------------------|--------------------------|
| $3^n$                        | $2^n \cdot n^{10}$ | <input type="checkbox"/> | <input type="checkbox"/> |
| $(\sqrt{n} + n)(30\sqrt{n})$ | $\sqrt{n}$         | <input type="checkbox"/> | <input type="checkbox"/> |

- c. Consider the recurrence relation  $T(n) = 4T(\sqrt[3]{n}) + O(n)$ . Can we solve this using the Master Theorem? **Explain** whether the structure of this recurrence fits the requirements of the Master Theorem. 02

CO2

- 2 a. Your friend gave you an integer (key) and asked you to find its integer square root without using built-in functions like `sqrt()` or exponentiation. For example, if the input is 49, the output should be 7 because  $7^2 = 49$ . If the square root is not an integer, return the floor value of the actual square root. For example, for an input of 26, the output should be 5.

Your friend's initial approach is to use linear search (time complexity  $O(n)$ ) as shown below.

```
def LinearSearchToFindSquareRoot(key):
    result = -1
    for i in range(1, key, 1):
        if i * i <= key:
            result = i
    return result
```

Now to find the square root, he is working on a possible solution **P**, which will result in time complexity  $O(\sqrt{n})$ . As an algorithm student, you propose another algorithm **R** within time complexity  $O(\log_2 N)$ , which is much faster than  $O(\sqrt{n})$

- i. Present the **solution your friend** was trying to achieve (**Algorithm P**) with pseudocode/programmable code/step-by-step logical instructions. 03
- ii. Present your proposed solution (**Algorithm R**) with pseudocode/programmable code/step-by-step logical instructions. 04

CO2

CO3

- 3 a. CO2 Imagine you're helping Dr. Eleanor, who's studying old papers about a long-gone city. Each paper talks about a year in the city's life. Some events brought prosperity to the civilization (*represented by a positive number*), while others brought decline and challenges (*represented by a negative number*).

Dr. Eleanor believes it is essential to study events in a continuous sequence to fully understand the rise and fall of civilizations. She can start with any manuscript, but once she begins, she must analyze the remaining manuscripts sequentially to maintain the timeline,

Given a list of numbers for each manuscript (indicating the prosperity or decline they brought), can you:

[2, -6, 8, 8, -9, -1, -2, 20]

- Tell Dr. Eleanor which group of years she should read about to learn of the city's best times. How good were those years combined? Show in detail how you found this out. 05
- Explain the time taken by your preferred method, particularly given the “ $n$ ” numbers of manuscripts Dr. Eleanor has in her studies. It should be an easy and quick method, as she has many papers to go through! 03

- 4 The “Road Transport and Highways Division” of Bangladesh is working on a project where they want to establish six-lane highways among the 8 divisions of Bangladesh. Currently, there are 9 highways connecting these 8 divisions as given in the map below.

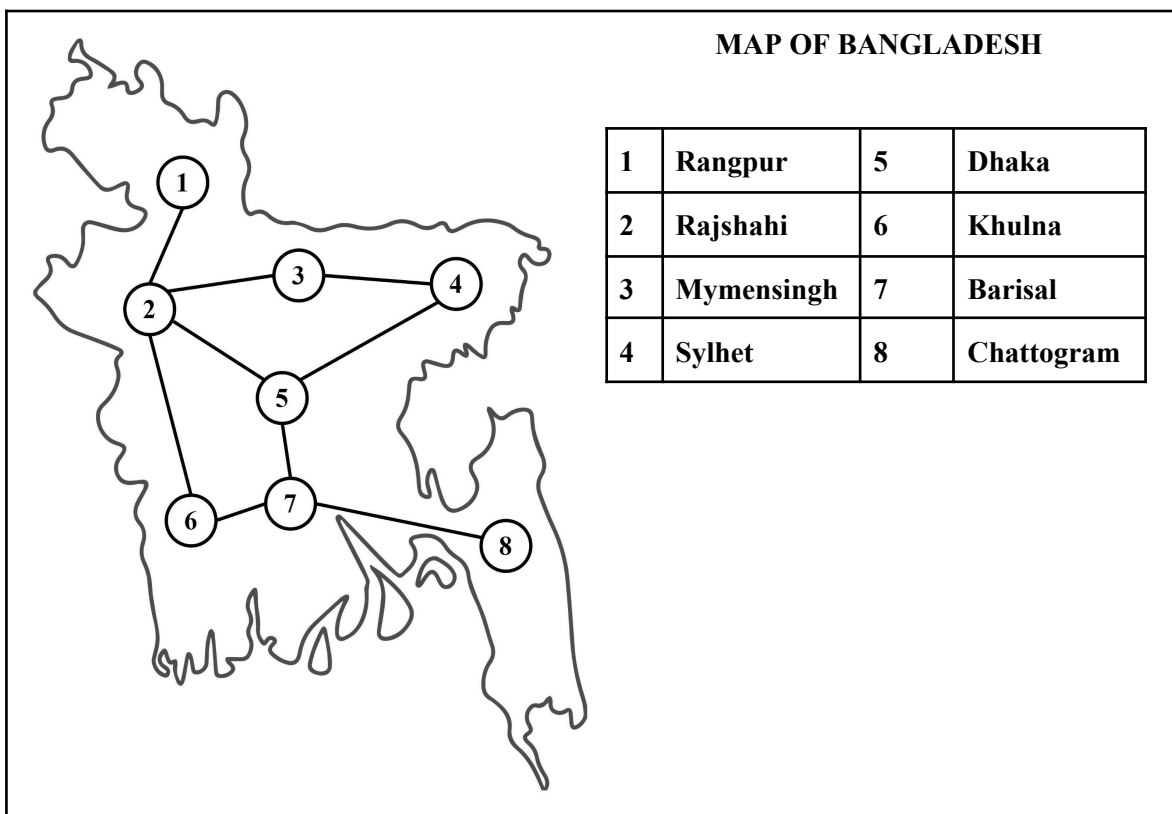


Fig 1: Graph For Question 4



**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Mid Semester Exam  
 Duration: 1 Hour 30 Minutes

Semester: Summer 2025  
 Full Marks: 30

**CSE 221: Algorithms**

Answer the following questions.  
 Figures in the right margin indicate marks.

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

- 1 a. Examine the following code segment and answer the following questions:  
 CO2

|                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>int funcB (int n) {     int m = 0;     for (int k = 1; k &lt;= n; k += 2) {         for (int j = n; j &gt;= 1; j--) {             for (int i = 1; i &lt;= n; i += 2) {                 if (i%2 != 0) {                     break;                 } else {                     m = k - j + i;                 }             }         }     }     return m; }</pre> | <pre>int funcA (int n) {     if (n == 1) {         return n;     } else {         x = funcA(n/2);         y = funcA(n/4);         z = funcA(n/6);         w = funcB(n/2);         return w;     } }</pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- (i) **Express** the time complexity of the function 'funcB' using the Big O-notation. 03  
 (ii) **Write** the recurrence relation of the function 'funcA'. (You can use your answer from question (i) if required.) 02  
 (iii) **Solve** the recurrence relation of no (ii) using any suitable method. 03

- b. **Prove** (with proper explanation or a formula):  $\sqrt[4]{10n^4 + 9n^3 + 8n} = \Theta(n)$  02  
 CO2

- 2 You are working with a weather monitoring station that records the temperature every hour for 12 hours during the night shift. During the night, the temperature gradually drops to reach the coldest hour, and then begins to rise again. Your task is to find the minimum temperature of the night, which corresponds to the coldest hour.

The station tracks temperature every hour over a 12-hour period. A sample pattern of the temperature reading per hour is given here:

|                  |    |    |    |    |    |    |   |    |    |    |    |    |
|------------------|----|----|----|----|----|----|---|----|----|----|----|----|
| Hour             | 1  | 2  | 3  | 4  | 5  | 6  | 7 | 8  | 9  | 10 | 11 | 12 |
| Temperature (°C) | 28 | 24 | 20 | 16 | 12 | 10 | 9 | 11 | 15 | 19 | 23 | 27 |

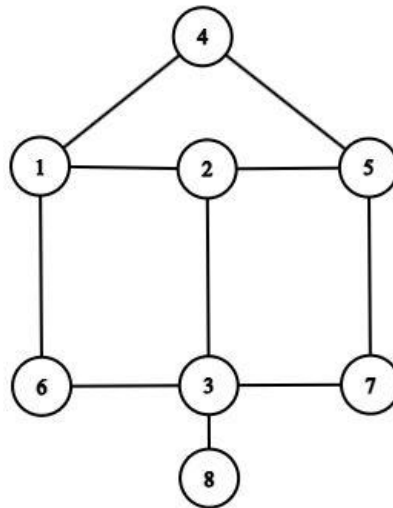
In this example, the temperature decreases from hour 1 to hour 7 (coldest at 9°C). Then it increases from hour 8 to hour 12.

- a. **Propose** a linear search approach to find the minimum temperature. Present the solution with pseudocode/programmable code/step-by-step logical instructions. **02**  
CO1
- b. Now **propose** a modification to binary search to solve the same problem. Present the solution with pseudocode/programmable code/step-by-step logical instructions. **03**  
CO3
- 3 a You are working on a grading system for a national standardized exam where students are marked out of 100. You need to sort millions of student scores as efficiently as possible. Speed is critical, and memory is not a major constraint. Comparisons are costly. **02**  
CO1
- Which algorithm is the most suitable for the above scenario? **Explain** your choice.  
**Write** the time complexity of your preferred algorithm.
- b Suppose your friend wants to implement a variant of Merge Sort called Ternary Merge Sort, where instead of dividing the array into two halves, **he divides it into three equal (or nearly equal) parts** and recursively sort each part, and then merge the three sorted subarrays into a single sorted array. It is guaranteed that the input array will have a size of at least 3. **02**  
CO3
- Now, he asks your help to **implement the ‘dividing step’ of the algorithm**. Present your idea with pseudocode/programmable-code/step-by-step logical instructions.
- c Does dividing into three subarrays make the algorithm asymptotically faster? **01**  
CO2 **Explain** why or why not.
- 4 You are participating in a competitive multiplayer strategy game where the objective is to defeat your opponent within a fixed time constraint. Throughout the game, various perks are provided at each level in the form of health boxes. At **Level 1**, you are presented with a list of health boxes, each represented by an integer. Positive values indicate a health boost to your own character, while negative values correspond to a reduction in your opponent's health. At this level, you are allowed to select exactly one box, and you want to maximize the health damage to your opponent. This list of health points is given below:
- [ 2, 3, -4, -3, 5, -1, 2, -2 ]
- Here, each element corresponds to a distinct health box.
- a For the above scenario, can you find the box that will cause the maximum possible reduction in enemy health by applying a **divide and conquer** approach? **Explain** your idea in brief, *no need to write full code*. **02**  
CO1

- b** At **Level 2**, you are offered the same collection of health boxes. However, this time you are allowed to choose **one or a block of consecutive boxes without gaps**. Now you want to increase your own health by collecting points that yield the highest points gain. You came up with an algorithm with  $O(N \cdot \log N)$  time complexity to compute this optimal health gain. Where 'N' is the number of health boxes. **03**
- CO3**

**Show** a simulation of the algorithm with proper steps.

- 5** Suppose, you and your classmates are trapped inside an ancient castle during a field trip. The only way out is to solve some tasks. The tasks involve navigating the castle and resolving ancient conflicts between the spirits that guard it. Your goal is to escape through the Gate of Reason before the castle resets and traps you forever. There are 8 rooms in the castle and you can navigate from one room to another using some corridors. The map of the castle is given below:



- a** Suppose, you are in Room 4. Your first task is reaching Room 6 which is the control room and the Gate of Reason is also situated here. However, the castle is flooding and some of the corridors have been blocked. How can you determine the minimum number of steps required to reach Room 6 from Room 4 avoiding the blocked corridors? **03**
- CO1**

**Explain** your idea in brief, *no need to write full code*.

- b** Now that you have reached Room 6 and stopped the flood, all corridors are accessible once again. Only one challenge remains: unlocking the Gate of Reason to escape the castle. Each of the 8 rooms in the castle is guarded by a magical spirit. The ancient Gate of Reason only opens if the spirits can be divided into two opposing groups such that:
- No two spirits in the same group are in conflict (the rooms they are guarding should not be connected by corridors).
  - The two groups must have the same number of spirits.
- CO1** **02**

**Propose** a suitable algorithm to determine whether the Gate of Reason can be opened. Present your idea with pseudocode/programmable-code/step-by-step logical instructions.

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**CSE 221: Algorithms**

Answer the following questions.  
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|-------|-----|----------|
| Name: | ID: | Section: |
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- 1 a. Examine the following code segment and answer the following questions:  
 CO2

|                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>int funcB (int n) {     int m = 0;     for (int k = 1; k &lt;= n; k += 3) {         for (int j = n; j &gt;= 1; j--) {             for (int i = 1; i &lt;= n; i += 2) {                 if (i%2 != 0) {                     break;                 } else {                     m = k - j + i;                 }             }         }     }     return m; }</pre> | <pre>int funcA (int n) {     if (n == 1) {         return n;     } else {         x = funcA(n/2);         y = funcA(n/4);         z = funcA(n/8);         w = funcB(n/4);         return w;     } }</pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- (i) **Express** the time complexity of the function 'funcB' using the Big O-notation. 03  
 (ii) **Write** the recurrence relation of the function 'funcA'. (You can use your answer from question (i) if required.) 02  
 (iii) **Solve** the recurrence relation of no (ii) using any suitable method. 03

- b. **Prove** (with proper explanation or a formula):  $\sqrt[4]{7n^4 + 6n^3 + 5n} = \Theta(n)$  02  
 CO2

- 2 You are analyzing the hourly shipment volumes of a logistics company. During the day, the shipment volume gradually increases to a peak hour, and then decreases. Your task is to find the hour with maximum shipment volume, i.e. the true peak hour.

The company tracks its shipment volume every hour over a 12-hour period. A sample pattern of the shipment volume is given here:

| Hour            | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |
|-----------------|---|---|---|---|----|----|----|----|----|----|----|----|
| Shipment volume | 2 | 4 | 6 | 8 | 10 | 13 | 17 | 15 | 12 | 7  | 4  | 1  |

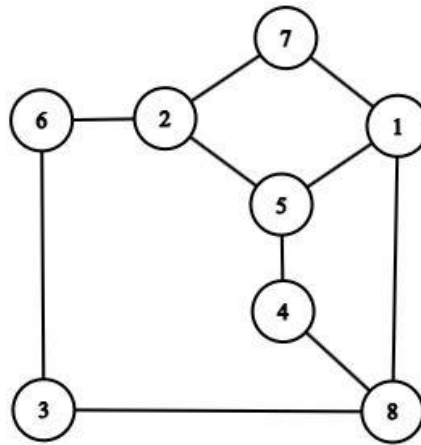
In this example, the volume increases from hour 1 to hour 7 (peak at 17). Then it decreases from hour 8 to hour 12.

- a. **Propose** a linear search approach to find the maximum hourly shipment volume. Present the solution with pseudocode/programmable code/step-by-step logical instructions. **02**  
CO1
- b. Now **propose** a modification to binary search to solve the same problem. Present the solution with pseudocode/programmable code/step-by-step logical instructions. **03**  
CO3
- 3 a You are developing an e-commerce platform that collects product ratings from users (out of five stars). For analytics, the system needs to sort millions of these ratings quickly every day. Memory is not an issue, but comparison operations are expensive. **02**  
CO1
- Which algorithm is the most suitable for the above scenario? **Explain** your choice.  
**Write** the time complexity of your preferred algorithm.
- b Suppose your friend wants to implement a variant of Merge Sort called Ternary Merge Sort, where instead of dividing the array into two halves, **he divides it into three equal (or nearly equal) parts** and recursively sort each part, and then merge the three sorted subarrays into a single sorted array. It is guaranteed that the input array will have a size of at least 3. **02**  
CO3
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- [-2, 3, 4, -3, -5, 1, 2, 2]
- Here, each element corresponds to a distinct health box.
- a For the above scenario, can you find the box that will cause the maximum possible reduction in enemy health by applying a **divide and conquer** approach? **Explain** your idea in brief, *no need to write full code*. **02**  
CO1

- b** At **Level 2**, you are offered the same collection of health boxes. However, this time you are allowed to choose **one or a block of consecutive boxes without gaps**. Now you want to increase your own health by collecting points that yield the highest points gain. You came up with an algorithm with  $O(N \cdot \log N)$  time complexity to compute this optimal health gain. Where 'N' is the number of health boxes. **03**
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**Explain** your idea in brief, *no need to write full code*.

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- CO1**
- No two spirits in the same group are in conflict (the rooms they are guarding should not be connected by corridors).
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